Mini Project 2
Schedule-Creating Agent

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Introduction

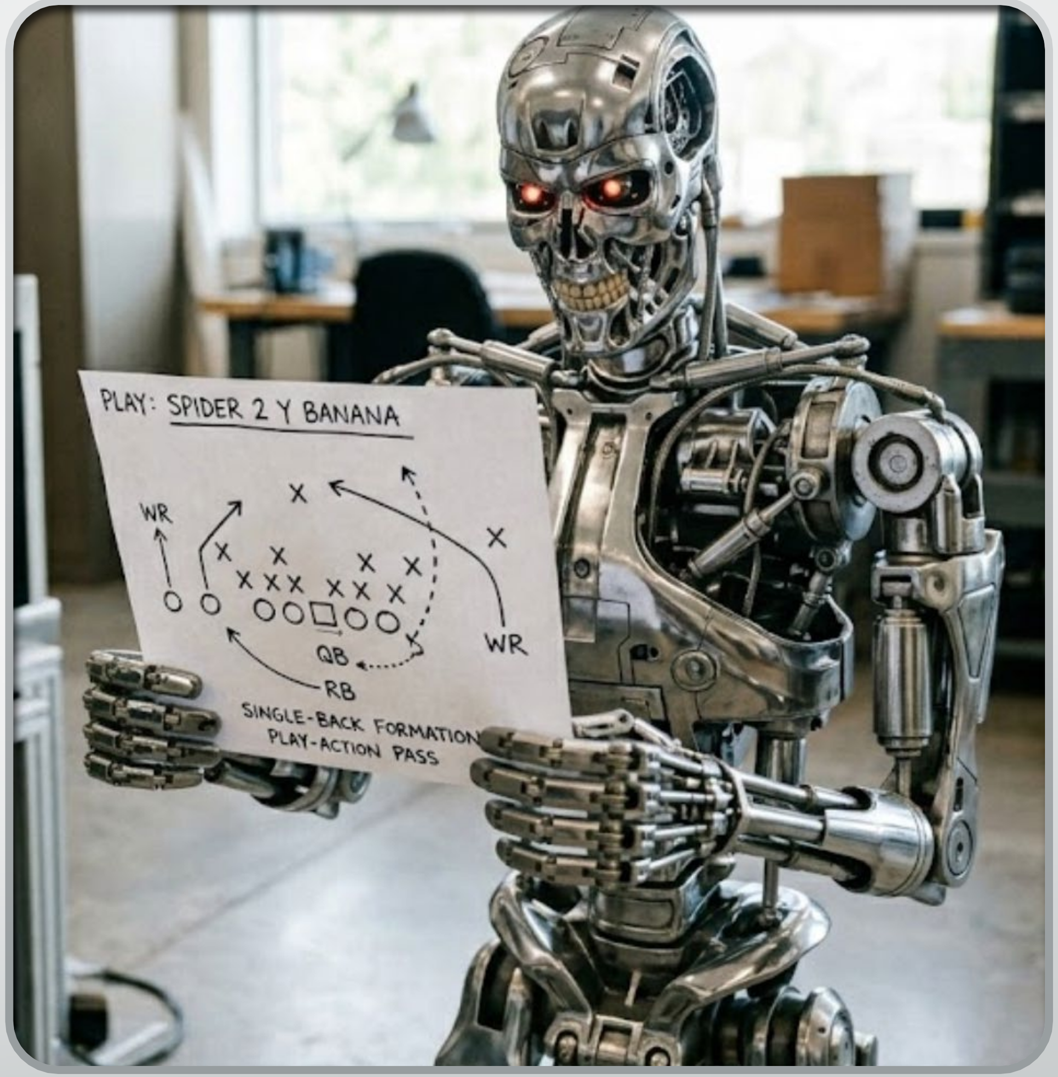
- Our team created an AI agent to generate effective study plans when it receives assignment PDFs as input
- Uses LangGraph so the agent can decide whether to use certain tools at each step

How PEAS was Implemented

- Performance measures:
  - Accuracy/usefulness of the generated study plan
  - Alignment with assignment deadlines and rubric requirements
  - Clarity of the output
- Environment:
  - Documents provided by the user
  - Local file system where input and output are stored
  - External knowledge accessed through the LLM and embedding models
- Actuators:
  - Ability to scan files
  - Produce textual responses
  - Generate excel spreadsheets
- Sensors:
  - PDF loaders
  - Vector databases for semantic search
  - User input prompts

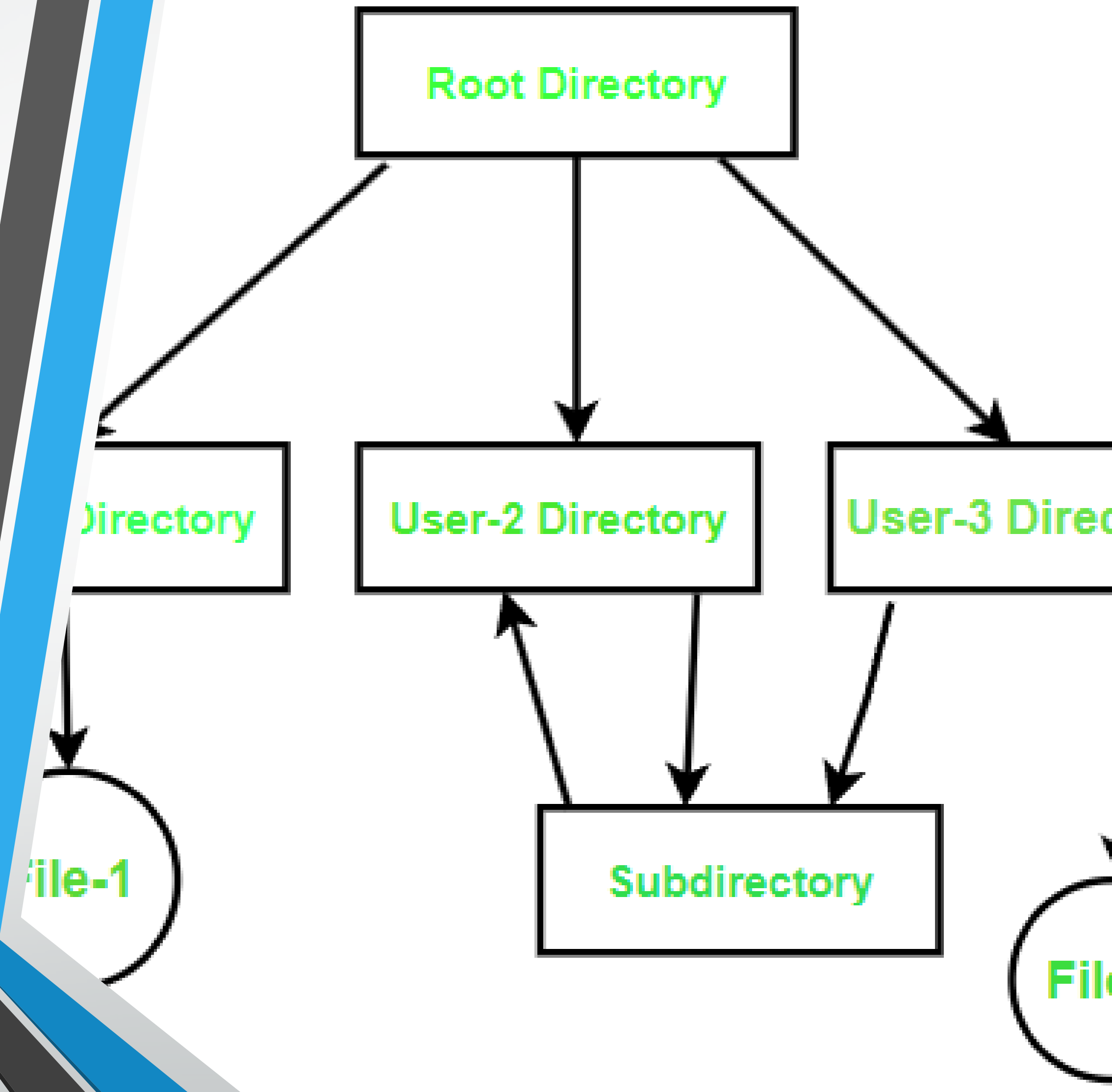
Approach: Plan-and-Execute

- Planner: The LLM first analyzes a user objective then breaks it down into a sequence of actionable sub-tasks
- Executor: takes individual steps from the plan and uses tools to perform actions
- Agent is partially observable because it relies on available information and must infer any missing parts



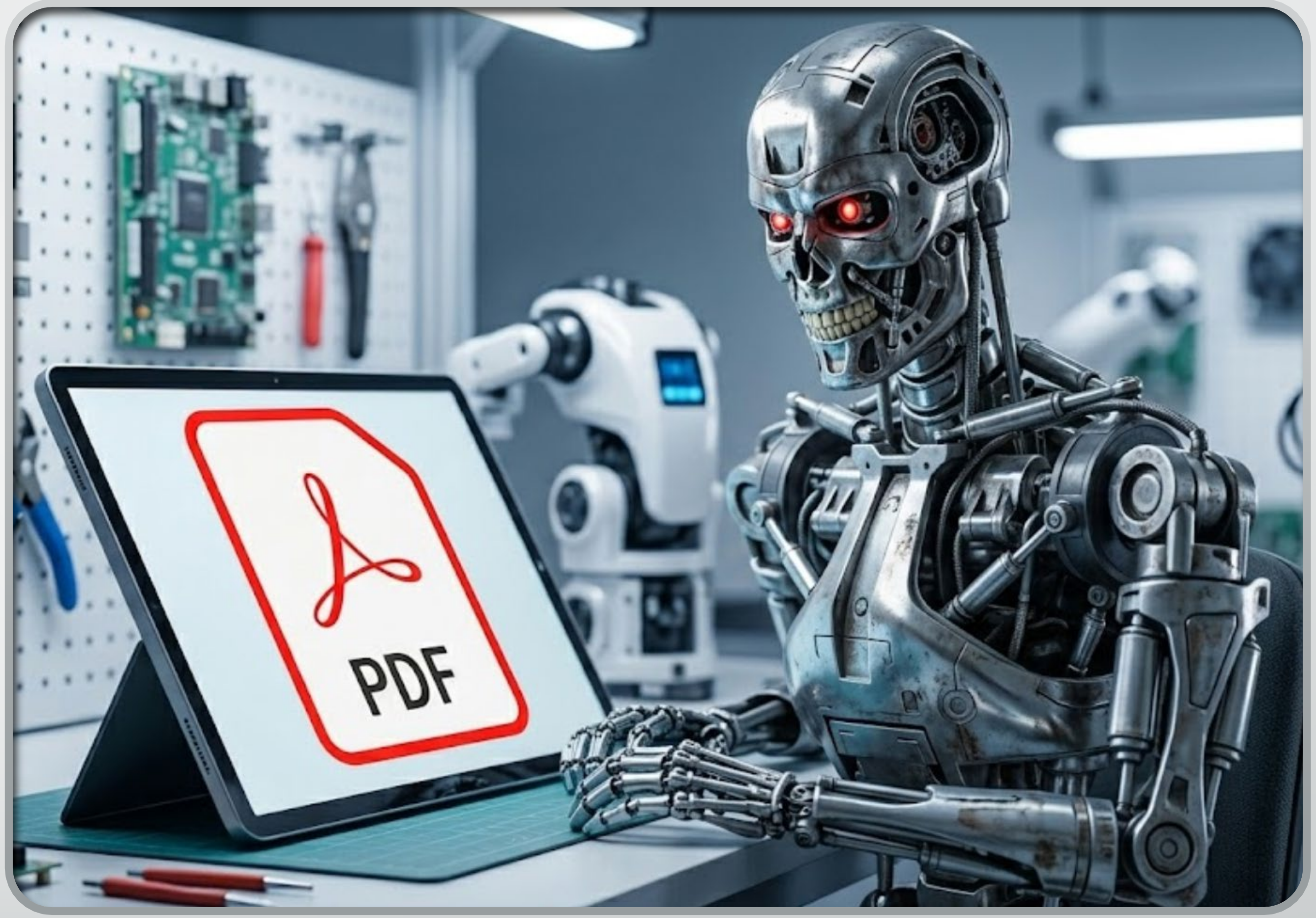
Directory Management Tool

- Uses the LangChain file_management toolkit
- Allows the agent to read / write files
- Navigates through directories



PDF Tool

- We globed every pdf on the directory
- Used PyPDFLoader to convert the pdf to markdown
- Using text embeddings, we stored each pdf in a single chroma vector database, allowing us to efficiently query the LLM about the contents
- We used the embeddings to call the LLM about the pdf contents



Excel Tool

- Form the GanttTask as a data structure for the input
- We used xlsxwriter to write the required columns one per item for the GnattTask
- Next, we used a bar chart which we modified to look like a Gnatt chart and rotated the y axis
- X-axis is the timeline and the Y-axis is the label

GnattTask: # remember this it will be referenced later

Title

Start date

End date

Requirements

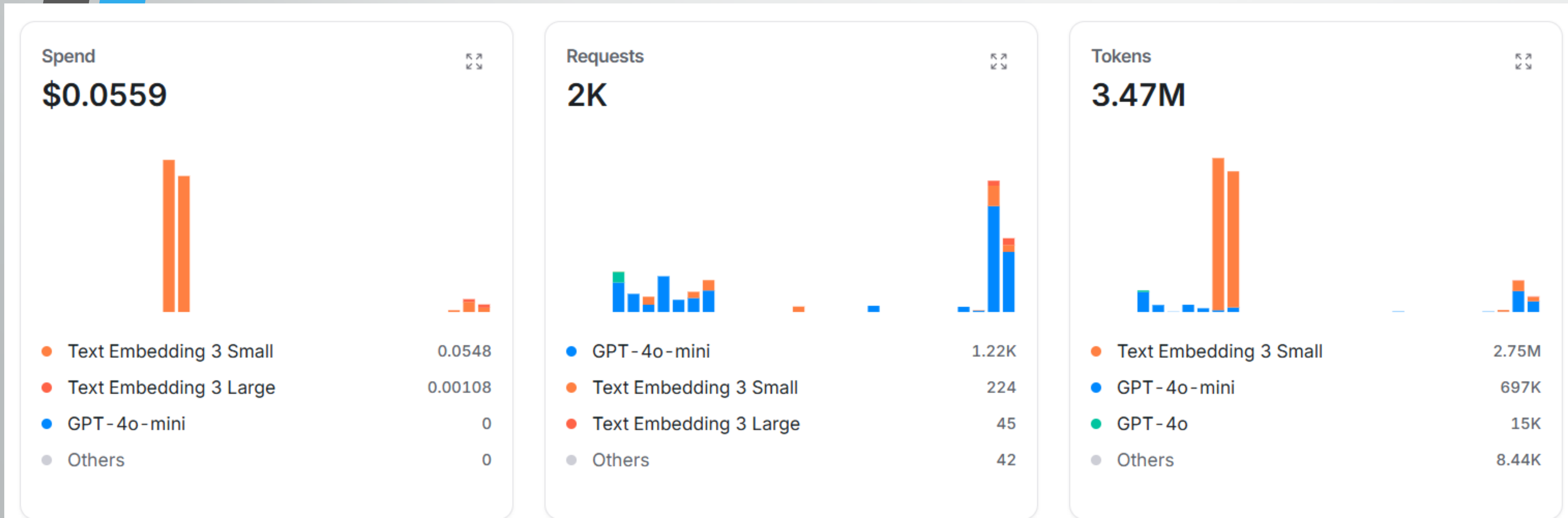
Priority

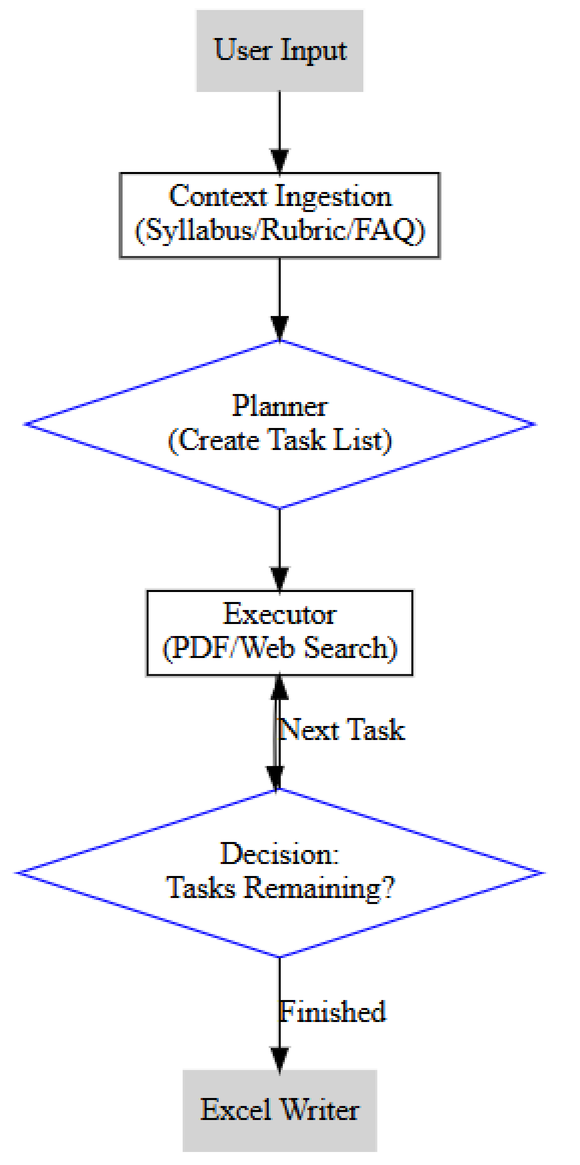
Days Needed

Notes

OpenRouter Models

- Allows us to use many different models ranging from Claude, Gemini and chatgpt. Essentially one place for all models. We put 2$ into it. Also, these credits don't expire unlike other platforms.
- Pro tip, don't leak your API key, use environmental variables
- Also don't tattoo your API key
- We went WILD 🤑 🤑 🤑 a whopping 5 cents mostly on the text embeddings.





AI Agent State and Nodes

AgentState:

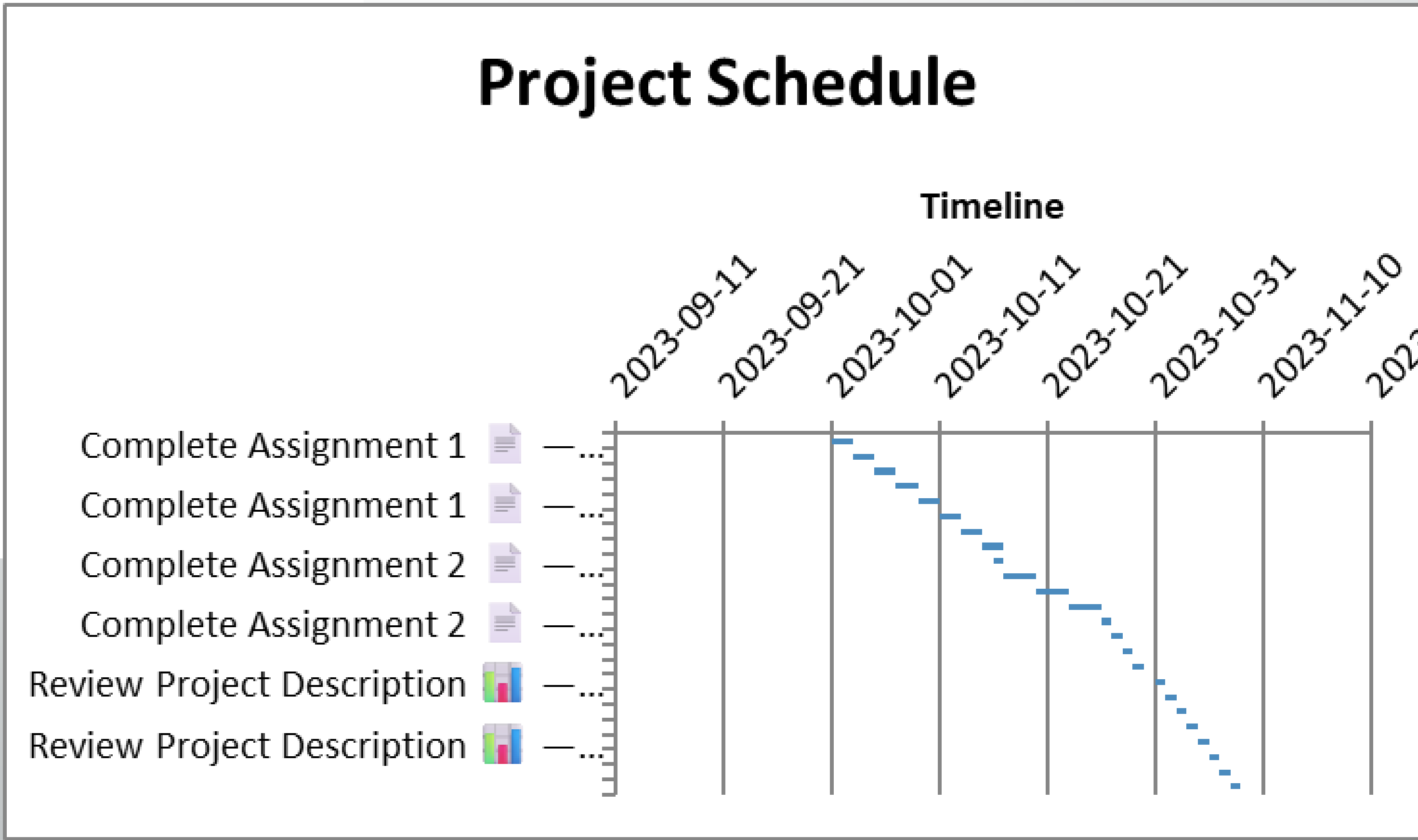
- Message aka user input
- List of rubric pdfs
- List of assignment pdfs
- current assignment index
- Summary of rubric, a string
- List of plans
- list of GanttTasks

Plan:

- title
- assignment path
- focus area
- start date
- due date
- notes

Result

```
Welcome to the LangGraph chatbot! Type 'quit', 'exit', or 'q' to stop.
User: Can you create a schedule for me. I am quite the procrastinator!!!
  Rubric files found: 1
  Assignment files found: 2

[PARSE NODE]
  Summary length: 1350 chars

[PLANNER NODE]
Plan items created: 2
  1. Assignment 1
  2. Assignment 2

[EXECUTOR NODE]
  Processing task 1: Assignment 1

[CONDITIONAL NEXT TASK NODE]

[EXECUTOR NODE]
  Processing task 2: Assignment 2

[CONDITIONAL NEXT TASK NODE]

[EXCEL WRITER NODE]
Gnatt Chart was created. End of agent.
```



Results part 2

Title	Start Date	Due Date	Requirements	Priority	Days	notes
Complete Assignment 1 📄 — Research Waterfall Model	2023-10-01	2023-10-03	Gather information on the advantages and limitations of the Waterfall mc	1	2	Use reliable sources and take notes! 📚
Complete Assignment 1 📄 — Draft Question 1	2023-10-03	2023-10-05	Write a draft response for Question 1 based on your research.	1	2	Focus on clarity and examples! ✍️
Complete Assignment 1 📄 — Research V-Model	2023-10-05	2023-10-07	Investigate how the V-Model addresses testing and validation compared t	1	2	Look for case studies or examples! 🔍
Complete Assignment 1 📄 — Draft Question 2	2023-10-07	2023-10-09	Write a draft response for Question 2 using your findings.	1	2	Make sure to evaluate effectiveness! 📝
Complete Assignment 1 📄 — Research Spiral Model	2023-10-09	2023-10-11	Explore how the Spiral model incorporates risk management into software	1	2	Find a relevant project example! 🏆
Complete Assignment 1 📄 — Draft Question 3	2023-10-11	2023-10-13	Write a draft response for Question 3 based on your research.	1	2	Justify your reasoning clearly! 💡
Complete Assignment 1 📄 — Research Scrum and Kanban	2023-10-13	2023-10-15	Compare and contrast Scrum and Kanban as Agile frameworks.	1	2	Look for strengths and weaknesses! ⚖️
Complete Assignment 1 📄 — Draft Question 4	2023-10-15	2023-10-15	Write a draft response for Question 4 using your findings.	1	2	Use examples to support your argument! 📊
Complete Assignment 2 📄 — Read Assignment Instructions 📖	2023-10-16	2023-10-17	Thoroughly read the assignment instructions and grading rubric.	2	1	Make sure to understand all requirements before starti
Complete Assignment 2 📄 — Identify Actors 👥	2023-10-17	2023-10-20	List all actors involved in the Library Resource Reservation System.	2	3	Think about who will use the system and their roles.
Complete Assignment 2 📄 — Identify Use Cases 📋	2023-10-20	2023-10-23	Draft a list of use cases based on the system description.	2	3	Focus on what actions users can perform.
Complete Assignment 2 📄 — Create Use Case Diagram 🖼️	2023-10-23	2023-10-26	Use a diagram tool to create the use case diagram including actors and us	2	3	Ensure clarity and accuracy in your diagram.
Complete Assignment 2 📄 — Incorporate Misuse Cases ⚠️	2023-10-26	2023-10-27	Identify at least two misuse cases and their mitigation strategies.	2	1	Think about potential security issues.
Complete Assignment 2 📄 — Review and Revise 🔍	2023-10-27	2023-10-28	Review the diagram and all components for accuracy and completeness.	2	1	Get feedback from peers if possible!
Complete Assignment 2 📄 — Prepare Submission Document 📄	2023-10-28	2023-10-29	Compile your work into a PDF or DOCX file, including your name and cours	2	1	Double-check formatting before submission!
Complete Assignment 2 📄 — Submit Assignment on Canvas 🚀	2023-10-29	2023-10-30	Upload your final document to Canvas before the deadline.	2	1	Set a reminder for the due date!
Review Project Description 📊 — Research Software Requirements	2023-10-31	2023-11-01	Gather information on software requirements for the project.	3	1	Start with online resources and previous projects. 📚
Review Project Description 📊 — Draft Software Requirements Spec	2023-11-01	2023-11-02	Create a draft of the SRS document based on your research.	3	1	Use clear and concise language. 📝
Review Project Description 📊 — Create Use Case Diagram	2023-11-02	2023-11-03	Design a use case diagram that outlines the system's functionality.	3	1	Utilize diagramming tools like Lucidchart or draw.io.
Review Project Description 📊 — Review and Revise SRS and Use Ca	2023-11-03	2023-11-04	Review the draft SRS and use case diagram, make necessary revisions.	3	1	Get feedback from peers if possible. 🔍
Review Project Description 📊 — Research Software Design and Arcl	2023-11-04	2023-11-05	Look into software design patterns and architecture styles relevant to you	3	1	Focus on what fits best for your project needs. 💡
Review Project Description 📊 — Draft Software Design and Architec	2023-11-05	2023-11-06	Create a draft document outlining the software design and architecture.	3	1	Be thorough and clear in your explanations. 📄
Review Project Description 📊 — Create UML Diagrams	2023-11-06	2023-11-07	Design UML diagrams that represent the system's structure and behavior.	3	1	Ensure diagrams are easy to understand. 📊
Review Project Description 📊 — Compile All Deliverables for Submis	2023-11-07	2023-11-07	Gather all documents and diagrams for submission.	3	1	Double-check everything before submission! ✅

On to the demo!